CANDIDATE'S NAME: ______________________________ **CTG:** _________

YISHUN JUNIOR COLLEGE

2011 JC2 PRELIMINARY EXAMINATION

CHEMISTRY
HIGHER 2

9647/01

Paper 1 Multiple Choice

FRIDAY 26 AUGUST 2011

1130hrs − 1230hrs

(1 hour)

Additional materials:

Multiple Choice Answer Sheet and Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your index number, name and CTG on the Answer Sheet.

There are forty questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider correct and shade your choice on the OMS provided.

1 *Use of the Data Booklet is relevant to this question*.

In leaded petrol, there is an additive composed of lead, carbon and hydrogen only. This compound contains 29.7% carbon and 6.19% hydrogen by mass.

What is the value of x in the empirical formula PbC_8H_x?

A 5 **B** 6 **C** 16 **D** 20

2 Naturally-occurring silicon is a mixture of three isotopes, ^{28}Si, ^{29}Si and ^{30}Si. The relative atomic mass of silicon is 28.109.

What could be the relative abundance of each of the three isotopes?

A 91.1% ^{28}Si, 7.9% ^{29}Si and 1.0% ^{30}Si

B 92.2% ^{28}Si, 4.7% ^{29}Si and 3.1% ^{30}Si

C 95.0% ^{28}Si, 3.2% ^{29}Si and 1.8% ^{30}Si

D 96.3% ^{28}Si, 0.3% ^{29}Si and 3.4% ^{30}Si

3 The table below gives the successive ionisation energies for an element ***A***.

	1st	2nd	3rd	4th	5th	6th
Ionisation energy / kJ mol^{-1}	950	1830	2770	4220	5810	12320

What could be the formula of the chloride of ***A***?

A $\boldsymbol{A}Cl$ **B** $\boldsymbol{A}Cl_2$ **C** $\boldsymbol{A}Cl_3$ **D** $\boldsymbol{A}Cl_4$

4 Silicon carbide (carborundum) is a shiny, hard, chemically inert material with a very high melting point. It can be used to sharpen knives and make crucibles.

Which type of structure explains these properties?

A a giant structure with covalent bonds between silicon and carbon atoms

B a giant structure containing ionic bonds between carbon and silicon ions

C a giant layer structure with covalent bonds between atoms and van der Waals' forces between the layers

D a simple molecular structure with strong covalent bonds between carbon and silicon atoms

5 In a calorimetric experiment, 1.60 g of a fuel is burnt. 45% of the energy released is absorbed by 200 g of water. The temperature of the water increases from 18 °C to 66 °C. Given that the specific heat capacity of water is 4.2 J g^{-1} K^{-1}, what is the total energy released per gram of fuel burnt?

A 25 200 J **B** 56 000 J **C** 89 600 J **D** 143 000 J

6 Some relevant redox half-equations are given in the table.

half-equation	$E^\ominus$ / V
$I_2(aq) + 2e^- \rightleftharpoons 2I^-(aq)$	+0.54
$2H^+(aq) + O_2(g) + 2e^- \rightleftharpoons H_2O_2(aq)$	+0.68
$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightleftharpoons 2H_2O(l)$	+1.77

What will be observed when a few drops of acidified aqueous hydrogen peroxide are added to an excess of aqueous potassium iodide?

A The solution turns brown and effervescence occurs.

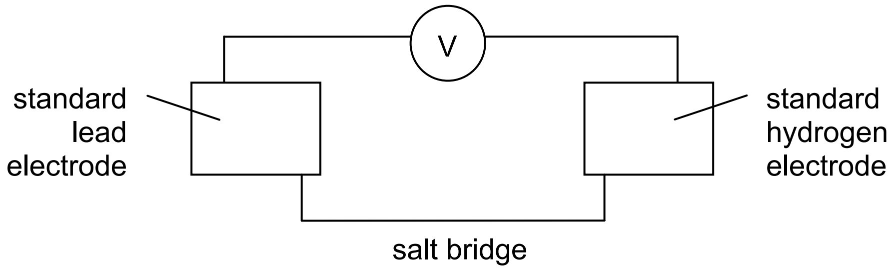
B The solution turns purple and effervescence occurs.

C The solution turns brown without effervescence.

D The solution turns purple without effervescence.

7 *Use of the Data Booklet is relevant to this question.*

The diagram represents an experiment to confirm the value of $E^{\ominus}(Pb^{2+}(aq)/Pb(s))$, the standard electrode potential of lead.



The e.m.f. of the cell was found to be 0.15 V rather than the expected 0.13 V.

Two students, ***J*** and ***K***, suggested possible explanations.

J: $[Pb^{2+}(aq)]$ was greater than 1.00 mol dm^{-3}
K: $[H^{+}(aq)]$ was greater than 1.00 mol dm^{-3}

Which of their suggestions could be correct?

A ***J*** only

B ***K*** only

C both ***J*** and ***K***

D neither ***J*** nor ***K***

8 Which of the following statements is correct about the following reaction?

$$\mathbf{P}(g) + \mathbf{Q}(g) \xrightleftharpoons{\mathbf{R}(s)} \mathbf{T}(g) \qquad \Delta H < 0$$

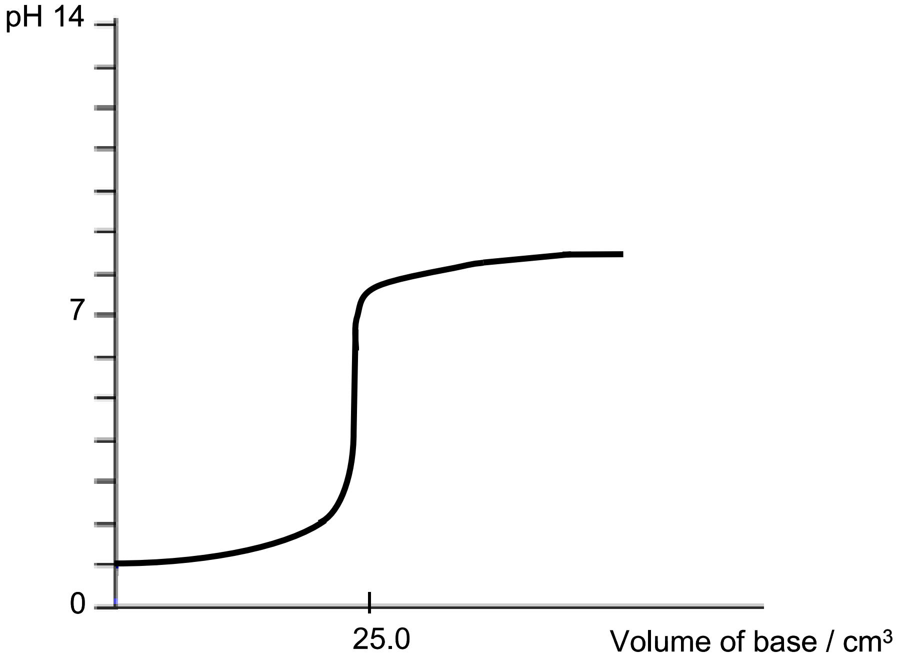
A The equilibrium constant will decrease with decreasing pressure.

B The equilibrium constant will increase with increasing temperature.

C The activation energy of the forward reaction is equal to that of the backward reaction.

D The solid **R** will lower the activation energy of both forward and backward reactions.

9 The graph shows the change in pH when an aqueous base is gradually added to 25 cm^3 of an acid.



Which of the following statements about the titration is correct?

A This is a titration between 0.10 mol dm^{-3} of HCl(aq) and 0.10 mol dm^{-3} of NH_3(aq).

B This is a titration between 0.10 mol dm^{-3} of benzoic acid and 0.10 mol dm^{-3} of NaOH(aq).

C Both phenolphthalein and methyl orange can be used as the indicator for this titration.

D Both acidic and basic buffers can be formed at different points during the titration.

10 The rate of removal of *Aspirin*, a pain-killer drug, from the body is a first order reaction with a rate constant, $k = 0.34\ h^{-1}$.

How long does it take for 87.5% of *Aspirin* to be removed from the body?

A 0.5 h **B** 1.0 h **C** 2.0 h **D** 6.1 h

11 Photochromic glass, used for sunglasses, darkens when exposed to bright light and becomes transparent again when the light is less bright. The depth of colour of the glass is related to the concentration of silver atoms.

The following reactions are involved.

reaction 1: $Ag^+ + Cl^- \rightleftharpoons Ag + Cl$

reaction 2: $Cu^+ + Cl \longrightarrow Cu^{2+} + Cl^-$

reaction 3: $Cu^{2+} + Ag \longrightarrow Cu^+ + Ag^+$

Which of the following statements about these reactions is correct?

A Light acts as an inhibitor in reaction 1.

B Light acts as a catalyst in reaction 2.

C Cu^+ ion is a homogeneous catalyst.

D Cu^+ ion is a heterogeneous catalyst.

12 Consecutive elements ***X***, ***Y***, and ***Z*** are in the third period of the Periodic Table. Element ***Y*** has the highest first ionisation energy and the lowest melting point.

What could be the identities of ***X***, ***Y***, and ***Z***?

A aluminium, silicon, phosphorus

B magnesium, aluminium, silicon

C silicon, phosphorus, sulfur

D sodium, magnesium, aluminium

13 Which property of Group II elements (beryllium to barium) decreases with increasing atomic number?

A reactivity with water

B second ionisation energy

C solubility of hydroxides

D thermal stability of carbonates

14 The reaction of 0.01 mole of chromium(III) chloride $[Cr(H_2O)_{6-x}Cl_x]Cl_y$ with excess $AgNO_3$ results in the precipitation of 2.87 g of AgCl.
Which one of the following information regarding the chromium complex is correct?

	x	y	**shape of complex**
A	1	2	octahedral
B	1	2	trigonal bipyramidal
C	2	1	octahedral
D	2	1	trigonal bipyramidal

15 The oxide of a metal **M** is insoluble in water but it dissolves in aqueous ammonia to form a complex cation $[\mathbf{M}(NH_3)_2]^+$. On reacting with dilute sulfuric acid, the metal oxide gives a deep blue solution and a deposit of a reddish brown metal, **M**.

What is the formula of the metal oxide?

A CuO **B** Cu_2O **C** Ag_2O **D** Fe_2O_3

16 Transition metals like platinum and rhodium are found in catalytic converters fitted on cars. Which of the following statements best explains the role of transition metals in this application?

A Transition metals can exhibit variable oxidation states in their compounds as the 3d and 4s electrons have similar energies.

B Transition metals have high melting points because both the 3d and 4s electrons are involved in forming strong metallic bonds.

C Transition metals have available and partially filled 3d orbitals for adsorption of reactant molecules.

D Transition metals formed coloured ions due to absorption of energy in the visible light region to promote an electron from a lower to a higher energy 3d orbital.

17 In which class of compound, in its general formula, is the ratio of hydrogen atoms to carbon atoms the highest?

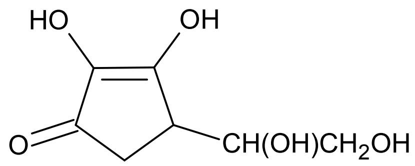
A alcohols

B arenes

C carboxylic acids

D halogenoalkanes

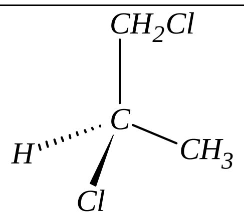
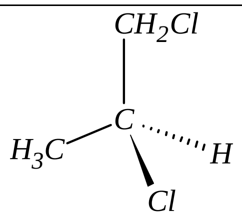
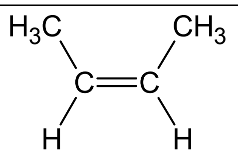
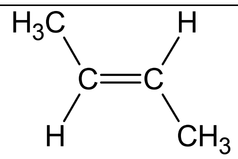
18 The diagram below shows the structure of Vitamin C.



How many chiral centres are there in one Vitamin C molecule?

A 0 **B** 1 **C** 2 **D** 3

19 In which pair is the melting point of isomer **II** expected to be higher than that of isomer **I**?

	isomer **I**	isomer **II**
A	$CH_3CH_2CH_2CH_2OH$	$(CH_3)_2CHCH_2OH$
B	$CH_3(CH_2)_4CH_3$	$(CH_3)_2CHCH(CH_3)_2$
C		
D		

20 Which of the following equations represents a valid propagation step in the free radical reaction between ethane and chlorine?

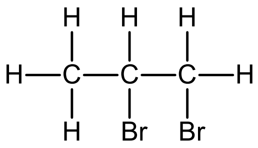
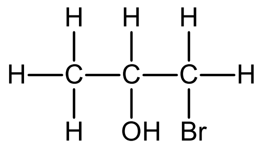
A $C_2H_6 + Cl\bullet \rightarrow C_2H_5Cl + H\bullet$

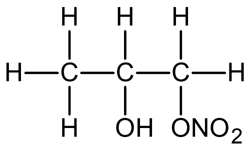
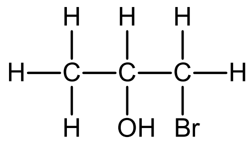
B $C_2H_5Cl + Cl\bullet \rightarrow C_2H_4Cl\bullet + HCl$

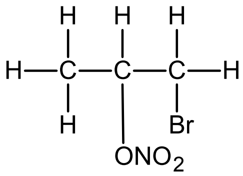
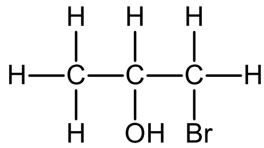
C $C_2H_5Cl + H\bullet \rightarrow C_2H_5\bullet + HCl$

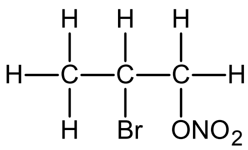
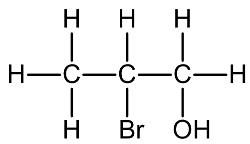
D $C_2H_5\bullet + Cl\bullet \rightarrow C_2H_5Cl$

21 When propene reacts with Br_2 in the presence of concentrated KNO_3, which are the two products that will be formed in greater proportion?

A  and 

B  and 

C  and 

D  and 

22 Four drops of 1-chlorobutane, 1-bromobutane and 1-iodobutane were put separately into three test-tubes containing 1.0 cm^3 of aqueous silver nitrate at 60 °C.

The following reaction occurred.

$$H_2O(l) + C_4H_9\boldsymbol{X}(l) + Ag^+(aq) \rightarrow C_4H_9OH(aq) + Ag\boldsymbol{X}(s) + H^+(aq)$$

[$\boldsymbol{X}$ = C*l*, Br, or I]

The rate of formation of cloudiness in the tubes was in the order $C_4H_9Cl < C_4H_9Br < C_4H_9I$.

Why is this so?

A The C-***X*** bond polarity decreases from C–C*l* to C–I.

B The solubility of Ag***X***(s) decreases from AgC*l* to AgI.

C The ionisation energy of the halogen decreases from C*l* to I.

D The bond energy of C-***X*** decreases from C−C*l* to C−I.

23 Chlorofluorocarbons (CFCs) have been widely used in aerosol sprays, refrigerators and in making foamed plastics, but are now known to destroy ozone in the upper atmosphere.

Which of the following will not destroy ozone, and therefore can be used a safer replacement for CFCs?

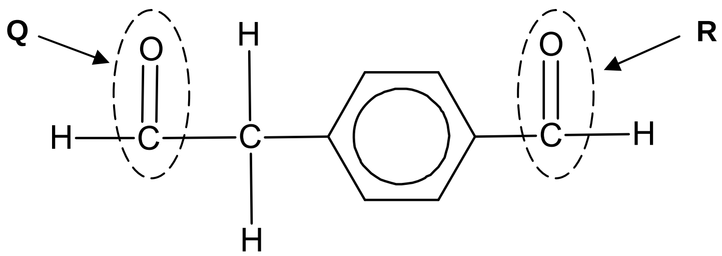
A $CH_3CH_2CH_2CH_3$

B CCl_3CBr_3

C $CHBr_3$

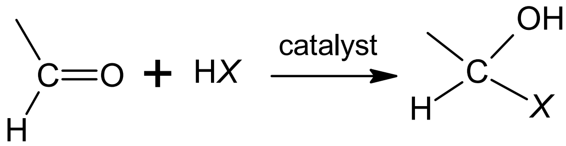
D $CHClFCClF_2$

24 When the following compound is tested in a laboratory with 2,4-dinitrophenylhydrazine and Fehling's reagent, which functional groups are responsible for positive tests?



	2,4-dintrophenylhydrazine	Fehling's reagent
A	**Q** and **R**	**Q** and **R**
B	**Q** only	**Q** and **R**
C	**Q** and **R**	**Q** only
D	**Q** and **R**	**R** only

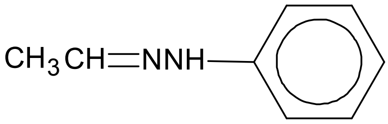
25 Aldehydes can undergo addition reactions with a variety of compounds of the form H*X* according to the following equation.



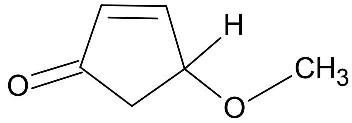
An example of such a reaction is the formation of cyanohydrin, where *X* = CN.

Which of the following compounds cannot be obtained by such an addition reaction to aldehyde, followed by dehydration?

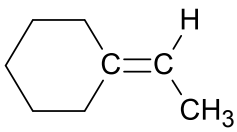
A



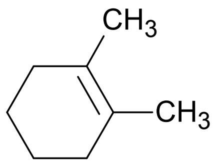
B



C



D



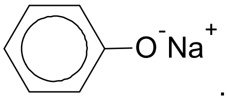
26 In a historically famous experiment, Wöhler heated 'inorganic' ammonium cyanate in the absence of air. The only product of the reaction was organic 'urea', $CO(NH_2)_2$. No other products were formed in the reaction.

What is the formula of the cyanate ion present in ammonium cyanate?

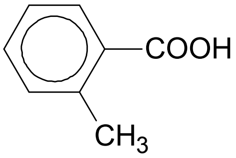
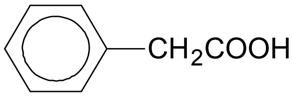
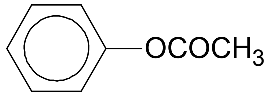
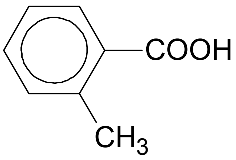
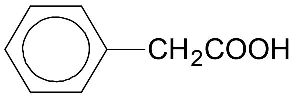
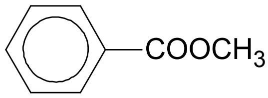
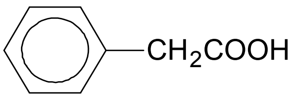
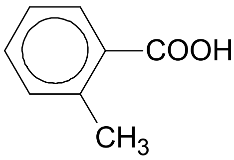
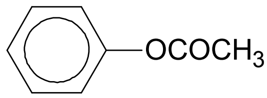
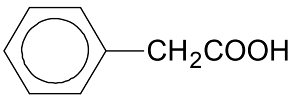
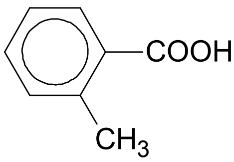
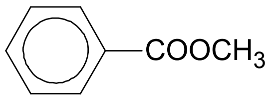
A CNO^- **B** CNO^{2-} **C** CO^- **D** NO^-

27 **E**, **F** and **G** are three isomeric aromatic compounds with the molecular formula, $C_8H_8O_2$. **E** and **F** are monobasic acids whereas **G** is a neutral compound.
The pK_a of **E** and **F** are 3.90 and 4.31 respectively.

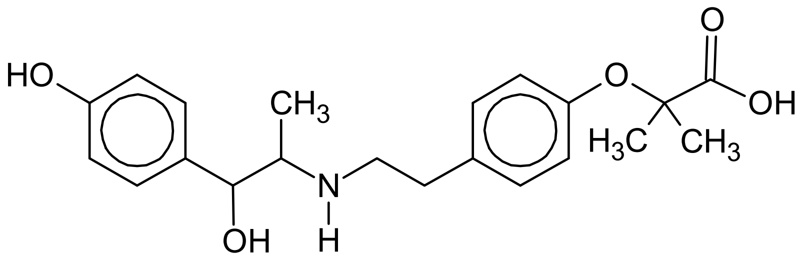
When **G** is heated strongly with NaOH, one of the organic products formed is:



Which of the following shows the correct identities of **E**, **F** and **G**?

	E	**F**	**G**
A			
B			
C			
D			

28 Compound **J** is used to treat urinary incontinence. The structure of **J** is shown below.



Which of the following statements about the reaction of compound **J** is correct?

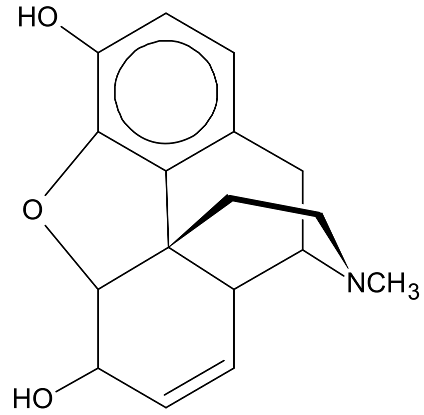
A 1 mole of **J** reacts with Na_2CO_3 to release 1 mole of CO_2.

B 1 moles of **J** reacts with Na metal to release 3 moles of H_2.

C 1 mole of **J** reacts with 2 moles of NaOH.

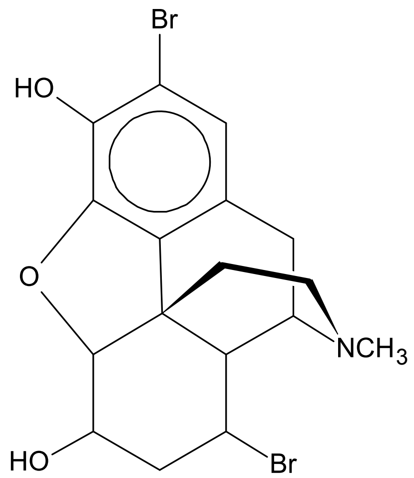
D 1 mole of **J** reacts with 3 moles of $SOCl_2$.

29 The structure of morphine is shown below.

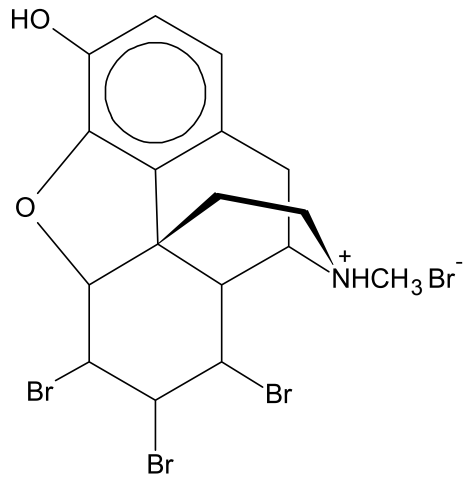


The reaction between morphine and dry HBr(g) gives

A



B



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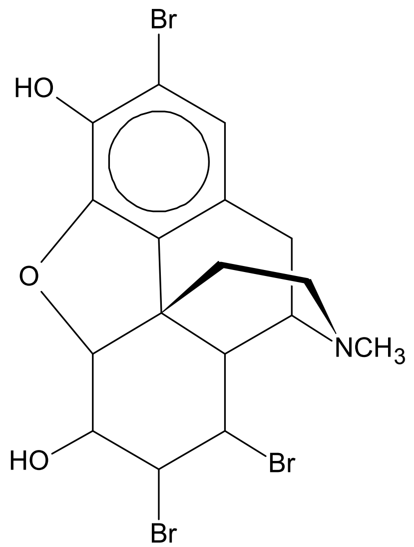
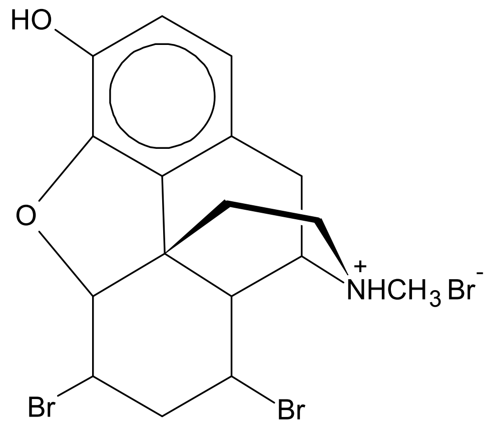
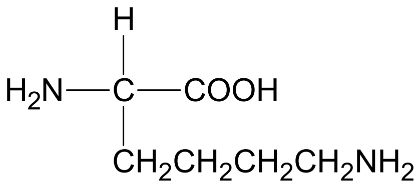


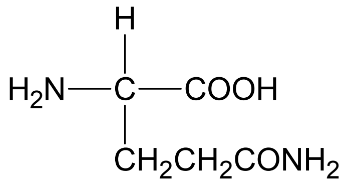
D



30 The amino acids, lysine and glutamine, can react with each other to form peptide linkages.



lysine



glutamine

What is the maximum number of different compounds, each containing one peptide linkage that can be formed from one molecule of lysine and one molecule of glutamine?

A 1 **B** 2 **C** 3 **D** 4

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

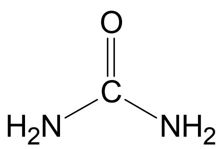
Decide whether each of the statements is or is not correct (you may find it helpful to pick a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

31 Which types of intermolecular forces can exist between adjacent urea molecules?



urea

1 induced dipole – induced dipole interactions

2 permanent dipole – permanent dipole interactions

3 hydrogen bonding

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

32 For which of the following reactions does the value of $\Delta H^{\ominus}$ represent both a standard enthalpy change of combustion and a standard enthalpy change of formation?

1 $C(s) + O_2(g) \rightarrow CO_2(g)$

2 $2C(s) + O_2(g) \rightarrow 2CO(g)$

3 $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$

33 *Use of the Data Booklet is relevant to this question.*

Which of the following are chemically stable when left to stand in the atmosphere?

1 a solution of potassium hexacyanoferrate(III)

2 a mixture of aqueous sodium hydroxide and iron(II) sulfate

3 a mixture of aqueous sodium chloride and vanadium(II) chloride

34 The table below shows the values of the ionic product of water, K_w, at two different temperatures.

Temperature / °C	K_w / $mol^2\ dm^{-3}$
25	1.00×10^{-14}
62	1.00×10^{-13}

What conclusions can be drawn from the K_w values?

1 At 62 °C, pH < 7

2 At 62 °C, pH = pOH

3 At 62 °C, pH = 14 – pOH

35 The chloride of element **L** is hydrolysed by water to form an acidic solution and its oxide reacts with acid to form a salt.

What could element **L** be?

1 magnesium

2 aluminium

3 silicon

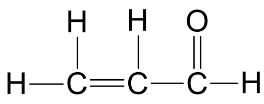
36 What happens when chlorine is bubbled through aqueous sodium hydroxide?

1 Disproportionation of chlorine occurs in both cold and hot sodium hydroxide solutions.

2 In cold sodium hydroxide solution, ClO_3^-(aq) ions are formed.

3 In hot sodium hydroxide solution, ClO^-(aq) ions are formed.

37 Acrolein is produced in photochemical smog. It has a strong smell, irritates eyes and mucous membrane, and is carcinogenic.



acrolein

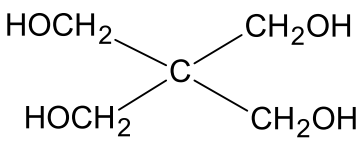
What can be deduced from the structure of acrolein?

1 All bond angles are approximately 120°.

2 It will undergo electrophilic addition.

3 It will undergo nucleophilic addition.

38 Pentaerythritol is an intermediate in the manufacture of paint.



Which of the following statements about pentaerythritol are correct?

1 It reacts with sodium.

2 It is dehydrated by concentrated sulfuric acid to an alkene.

3 Its empirical formula is CH_3O.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

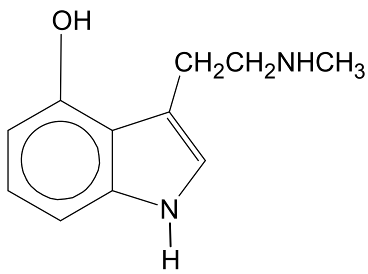
39 Which products are formed when phenylpropanoate undergoes acidic hydrolysis in the presence of water labeled with $^{18}\mathbf{O}$ isotope?

1 $CH_3CH_2CO^{18}\mathbf{O}H$

2 $C_6H_5{}^{18}\mathbf{O}H$

3 $C_6H_5CO^{18}\mathbf{O}H$

40 Psilocin is a psychedelic mushroom alkaloid which is the active compound that produces hallucinations from ingesting "magic mushrooms" and amplifies sensory experience. Compound **W** is a derivative of Psilocin.



compound **W**

Which of the following statements about compound **W** are correct?

1 The nitrogen-containing group in the ring has a lower pK_b than the nitrogen-containing group in the side chain.

2 It reacts with both aqueous acids and alkalis.

3 It gives white fumes with CH_3COCl.

~ END OF PAPER 1~